ConvDeck: Conversational Paper-to-Slide Generation via Stage-Specific User Feedback

Tarik Can Ozden, Sachidanand VS, Furkan Horoz, Ozgur Kara, Dilek Hakkani-Tür, Junho Kim, James M. Rehg

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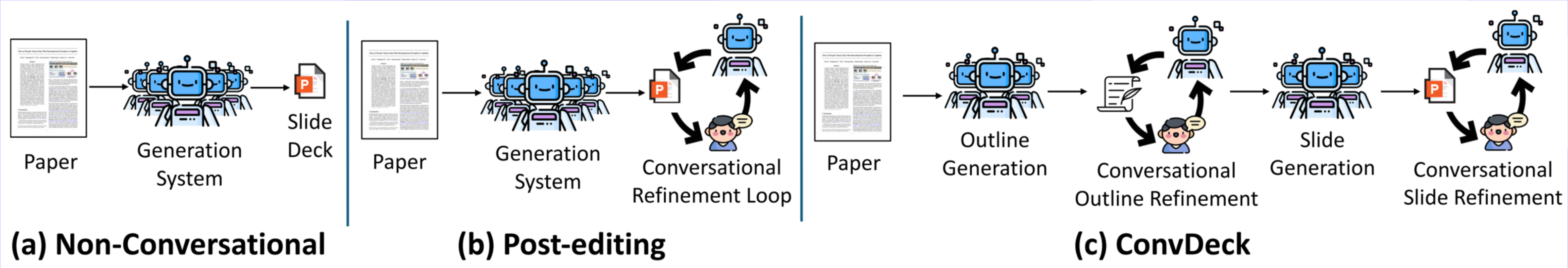
Motivation & Gap: From Single-Pass Slides to Conversational Control



- Paper-to-slide generation demands adaptable narratives and emphasis for diverse audiences.
- Single-pass and post-hoc edits leave structural issues unresolved early in the pipeline.
- Internal critics miss user-specific goals and presentation context during planning.

- Multi-stage design enables earlier intervention on outline and content allocation.
- Conversational control targets better alignment with **presenter goals**.

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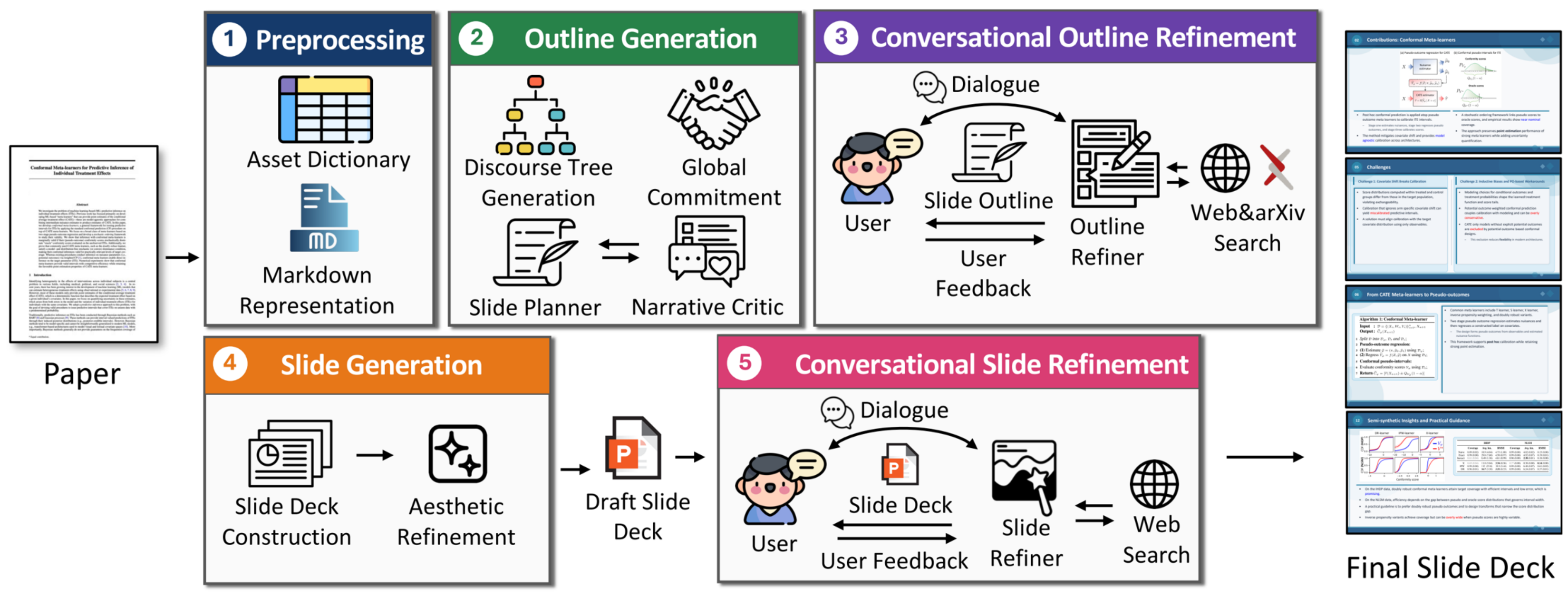
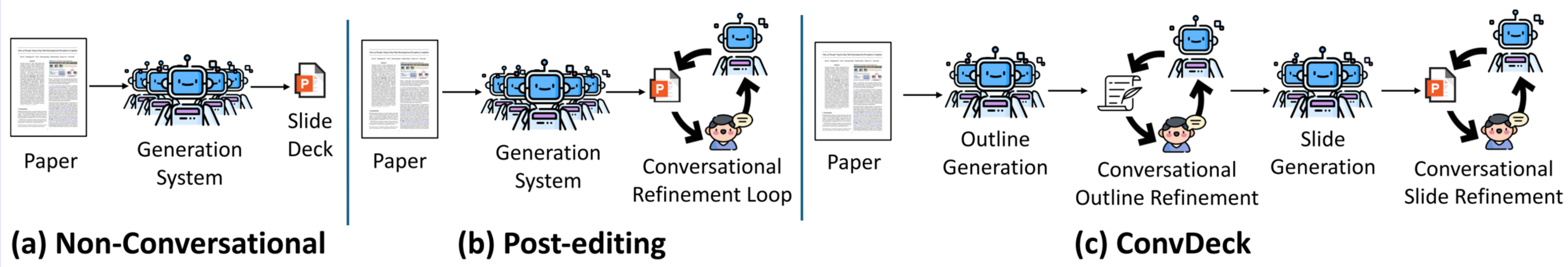
Introducing ConvDeck: Multi-Agent, Stage-Specific Conversational Refinement



- Two conversational loops operate at outline refinement and slide refinement stages.
- Agents reason, ask, and edit using a think, speak, and act protocol.
- Evaluation covers slide quality and user-goal satisfaction without sacrificing coherence.

- Stage-aligned feedback enables fine-grained control with **localized edits**.
- Structured interactions improve goal fulfillment across diverse categories.

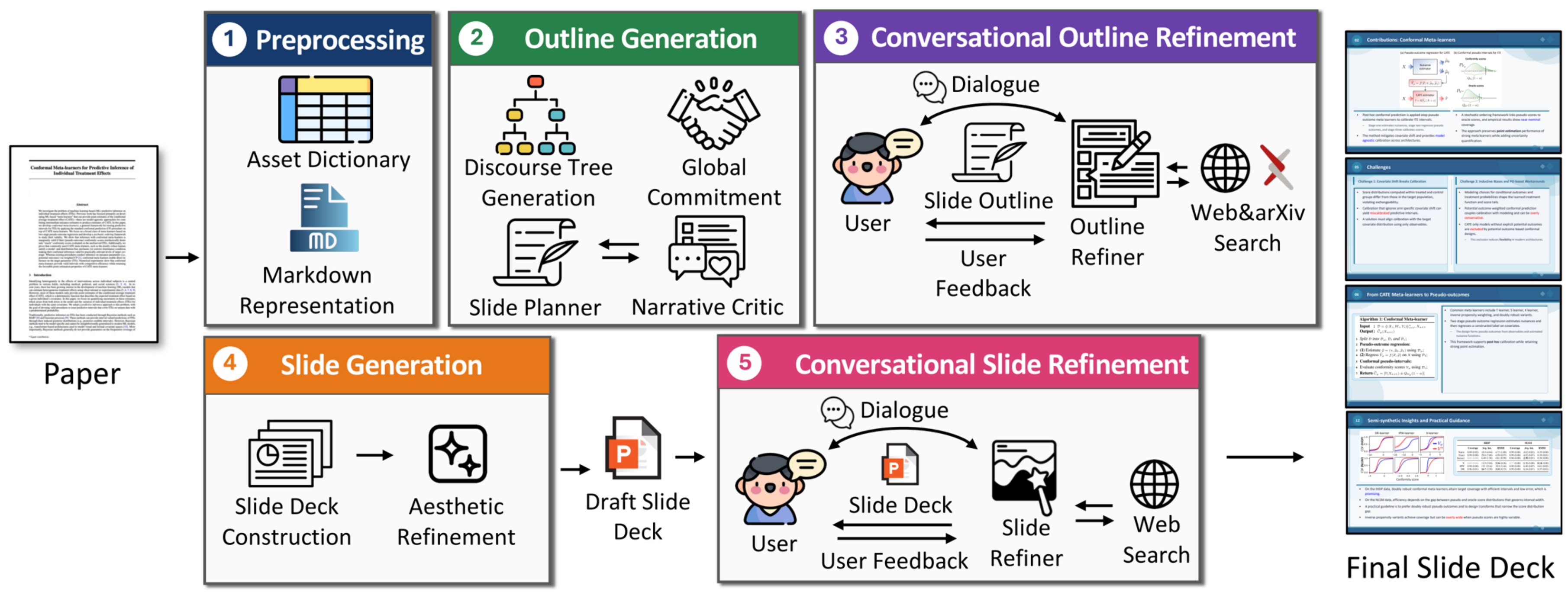
03 Related Work: Agentic Pipelines vs. Post-hoc Conversational Editing



- Prior systems span non-interactive pipelines, post-hoc editors, and personalization methods.
- Template/reference-driven; narrative/discourse-centered; critique–revise loops; personalization; posters/videos.
- ConvDeck uniquely routes feedback to the most effective stage for impact.

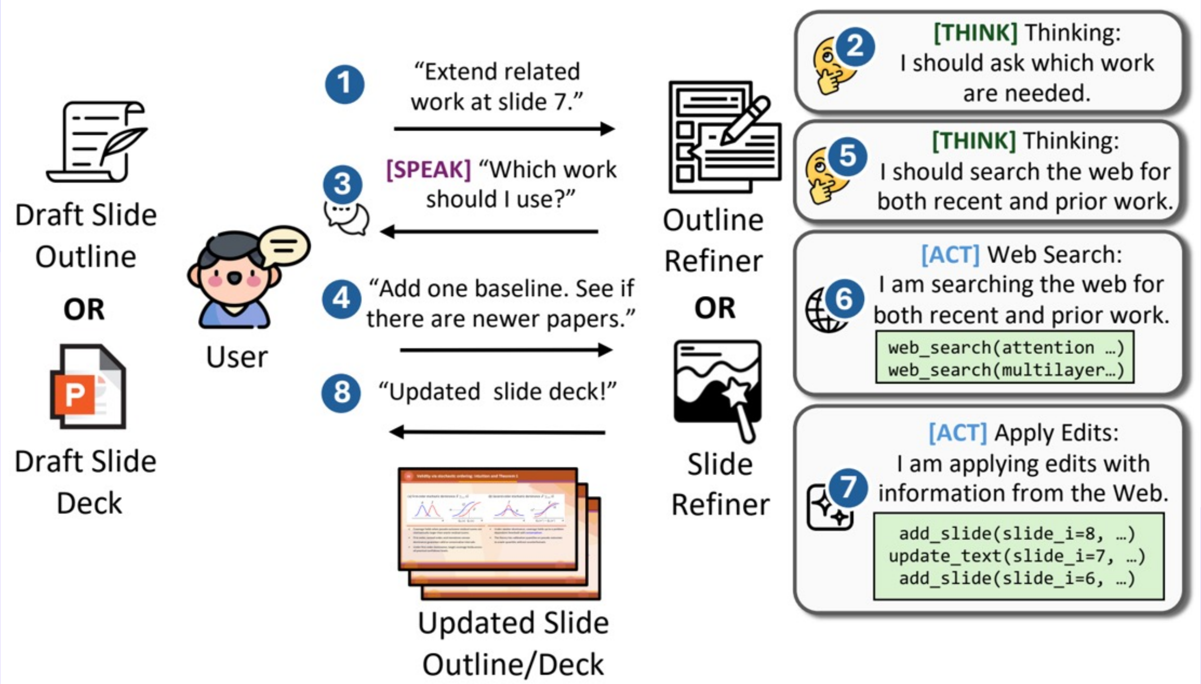
- Our positioning emphasizes stage-aligned loops and **coherent refinement**.
- Evaluation extends beyond quality to **user-goal satisfaction** and alignment.
- Conversational editing after rendering often shows limited structural leverage.

04 System & Pipeline: Five Stages + Key Components



- Five stages progress from preprocessing to conversational slide refinement with clear outputs.
- Structured JSON paired with PPTXGenJS enables precise and reversible edits.
- Outline interaction precedes rendering to shape narrative and coverage effectively.
- Slide interaction polishes content, layout, and assets after materialization.
- Two refinement loops support stable iteration with **low edit cost**.

05 Mechanisms: Think-Speak-Act and Local Edit Operations



- Outline operations include add, edit, split, merge, remove, and reorder with optional retrieval.
- Slide operations include insert, delete, modify, layout adjustments, and typography tuning.
- Think–Speak–Act enables clarification or immediate local edits.
- Rendered previews guide fixes for **overflow** and l**ayout crowding**.
- Structured edits maintain revision stability across refinement rounds.

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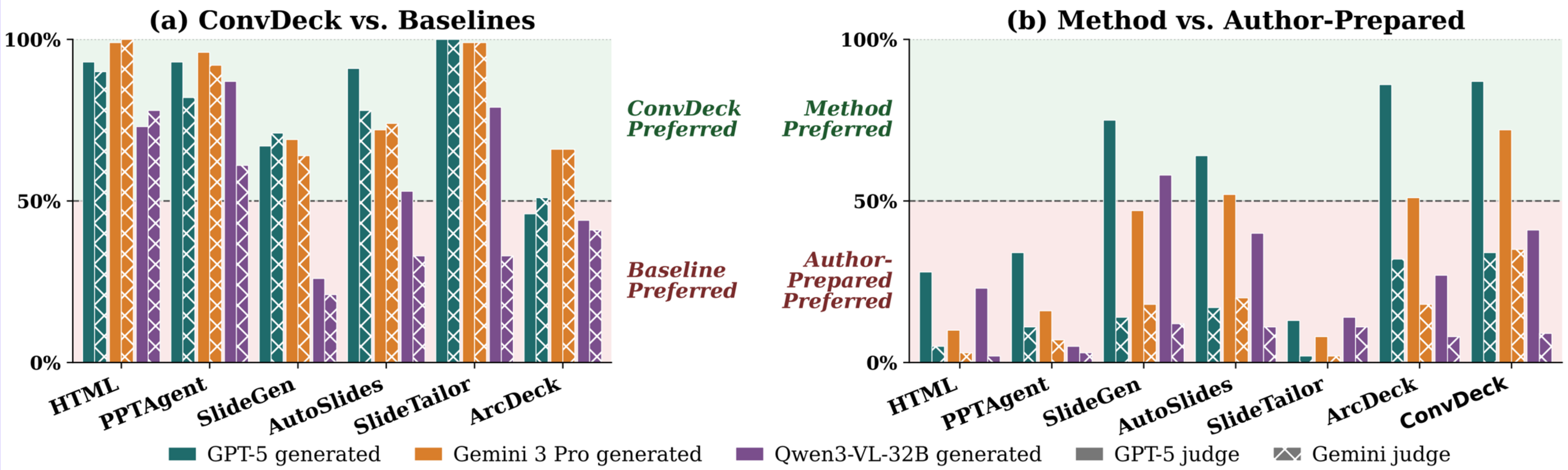
Baselines, Dataset, and Protocol



- Comparisons include non-conversational and post-hoc conversational baselines on ArcBench.
- ArcBench provides 100 paper and slide pairs with cross-judge evaluation to limit bias.
- Up to five refiner and user rounds proceed with early stopping for efficiency.
- Multiple backbones and cross-judging reduce self-preference bias.
- Success rate measures the fraction of runs that render a complete deck.
- Baselines: HTML, PPTAgent, SlideGen, SlideTailor, ArcDeck; conversational: ConvHTML, AutoSlides.

07 Evaluating User-Goal Satisfaction: Goals, Categories, and Alignment

Group	Category	Abbr.
Outline-relevant ($\mathcal{C}_{\text{out}}$) **Conv. Outline Refinement** (Stage 3)	Content Inclusion/Exclusion	Content
	Narrative Structure	Narrative
	Deck Composition	Composition
Slide-relevant ($\mathcal{C}_{\text{sld}}$) **Conv. Slide Refinement** (Stage 5)	Figure/Table Usage	Fig/Table
	Style & Wording	Style

- Stage-aligned refinement is hypothesized to better meet presenter-specific requirements.
- Goals map to outline categories and slide categories for targeted evaluation.
- Evaluation emphasizes **verifiable goals** and clear alignment signals.

Study 1: Does Conversational Refinement Close Unmet Goals?

Method	Content				Narrative				Fig/Table				Style				Composition				Overall				Success Rate		
Eval. models →	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*	User	GPT-5	GMN	*Avg*
ConvHTML	62%	68%	65%	80%	68%	72%	70%	72%	66%	76%	71%	70%	62%	76%	69%	64%	40%	66%	53%	74%	60%	72%	66%	72%	90%	100%	95%
AutoSlides	76%	64%	70%	77%	56%	68%	62%	72%	84%	72%	78%	65%	42%	48%	45%	77%	98%	86%	92%	77%	71%	68%	70%	74%	72%	91%	82%
ConvDeck	92%	88%	90%	81%	82%	82%	82%	75%	76%	88%	82%	74%	50%	66%	58%	81%	72%	96%	84%	78%	74%	84%	79%	78%	99%	96%	98%

- ConvDeck satisfies 79%, versus ConvHTML at 72% and AutoSlides at 70%.
- Performance is strong on content, narrative, figures and tables, and composition while style remains harder.
- Success rate remains high across backbones with stable rendering outcomes.
- Category gains reflect propagated fixes from outline into final slides.
- Residual style gaps indicate surface-level challenges under constraints.

Stage-wise Gains: Outline vs. Slide Refinement

Category	Outline (Stage 3)			Slide (Stage 5)		
	Before	After	Δ	Before	After	Δ
Content	42%	91%	↑ +49%	90%	93%	↑ +3%
Narrative	18%	89%	↑ +71%	66%	72%	↑ +6%
Composition	53%	96%	↑ +43%	80%	96%	↑ +16%
Fig/Table	–	–	–	54%	90%	↑ +36%
Style	–	–	–	52%	81%	↑ +29%
Avg.	37.7%	92.0%	↑ +54.3%	68.4%	86.4%	↑ +18.0%

- Outline refinement delivers large improvements in narrative and content planning.
- Slide refinement boosts presentation aspects including figures, tables, and style.
- Complementary stages combine structural gains with **visual polish**.
- Composition benefits appear at both outline and slide phases.
- Targeted loops reduce misplaced edits that undermine coherence.
- Narrative +71, Content +49, Fig/Table +36, Style +29.

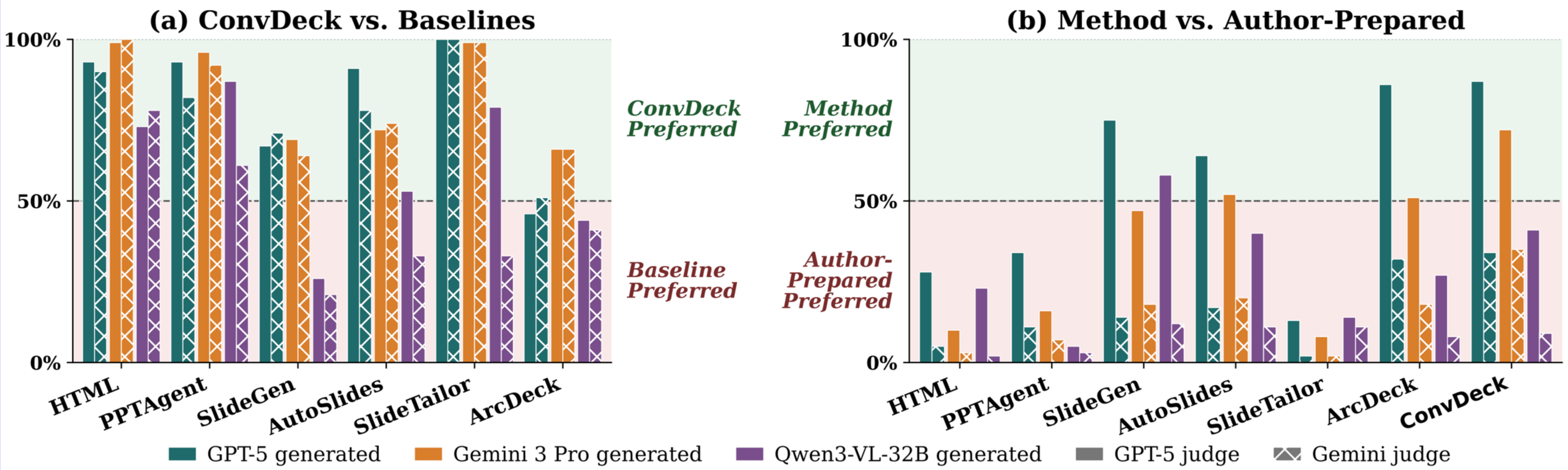
Round Dynamics: Rapid Early Gains and Stability

Metric	Outline generation (Stage 3)						Slide generation (Stage 5)					
	R0	**R1**	**R2**	**R3**	**R4**	**R5**	**R0**	**R1**	**R2**	**R3**	**R4**	**R5**
Sat. (%)	12.3	72.0	78.0	**83.3**	83.0	82.8	49.0	77.4	82.4	82.6	**84.2**	82.1
Sat. Δ	–	+59.7	+65.7	+71.0	+70.7	+70.5	–	+28.4	+33.4	+33.6	+35.2	+33.1
FU	–	4.83	4.80	4.78	4.58	4.42	–	4.61	4.47	4.51	4.24	3.74
FR	–	4.76	4.74	4.72	4.54	4.26	–	4.23	4.21	4.19	4.02	3.41
CRC	–	4.93	4.88	4.90	4.82	4.83	–	5.00	4.72	4.72	4.61	4.47
RS	–	4.89	4.85	4.89	4.83	4.76	–	4.66	4.56	4.56	4.42	4.21
Avg.	–	4.85	4.82	4.82	4.69	4.57	–	4.63	4.49	4.50	4.32	3.96
Avg. Δ	–	–	−0.03	−0.03	−0.16	−0.28	–	–	−0.14	−0.13	−0.31	−0.67

- Early rounds achieve most gains with diminishing returns in later interaction.
- High consistency and stability persist across consecutive rounds of refinement.
- Conversation quality remains strong despite slight later-round decline.

- Front-loaded progress provides efficient convergence on goals.
- Extended rounds can introduce minor drift without added benefit.

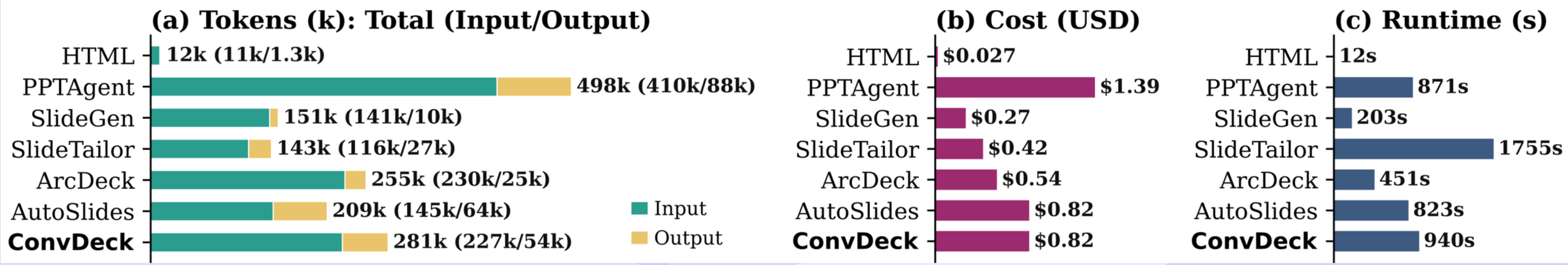
Overall Quality: Preference vs. Baselines and Author-Prepared



- ConvDeck is preferred over baselines with strong backbones under cross-judging.
- Quality approaches author-prepared decks while weaker backbones converge near parity.
- Backbone strength modulates preference with larger margins on top-tier models.

- Pairwise wins signal robust narrative and **clear visuals**.
- Judging artifacts may cause reference bias against generated decks.

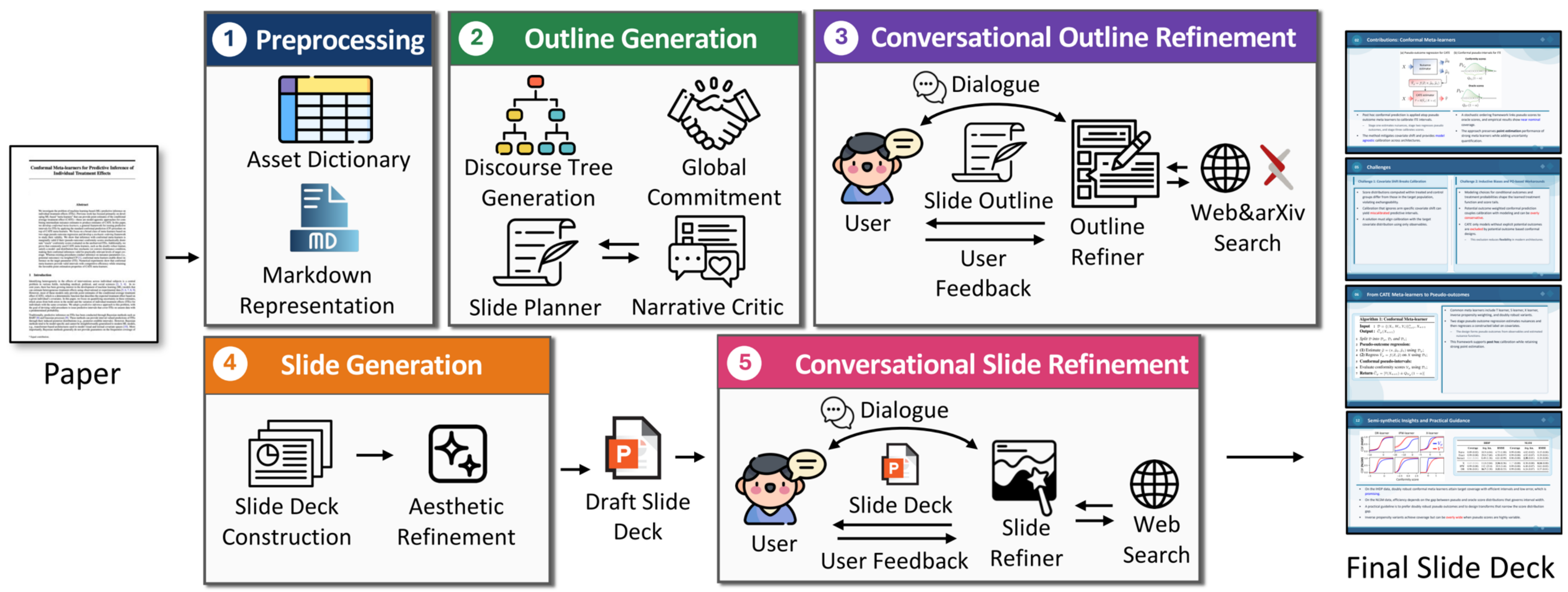
Compute Footprint: Cost and Runtime



- Token cost remains moderate and competitive with comparable baselines.
- Runtime overhead stays modest relative to added interactive capability.
- Structured JSON enables low-cost edits across iterative rounds.

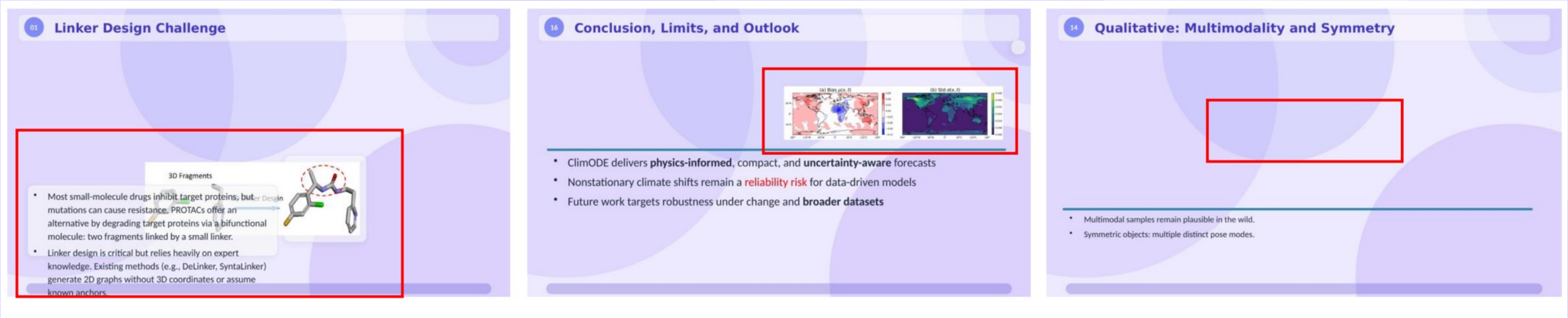
- Overall footprint reflects balanced tradeoffs between quality and speed.
- Repeated reads introduce overhead mitigated by localized changes.
- Tokens per paper: 281k; cost approx $0.82; runtime 940 s; ArcDeck 451 s

Conclusion: Stage-Aligned Conversation Improves Slide Generation



- Two-loop conversational design improves user-goal satisfaction over post-hoc editing.
- Quality and cost-effectiveness are maintained across backbones and settings.
- Stage-specific refinements provide coherent decks with **targeted changes**.
- Outline-first control propagates structural improvements through rendering.
- Slide-level polishing finalizes clarity, layout, and asset choices.

Limitations & Scope



(a) Overflow (b) Wrong Placement (c) No-Image

- Heavy reliance on an LLM simulator requires broader human evaluation for robustness.
- Layout sensitivity and backbone dependence reduce consistency on weaker models.
- Prototype state needs engineering hardening before production deployment.

- Visual inspection may miss rare overflow or crowding edge cases.
- Open-model gaps highlight performance variance across settings.

Thank You

Questions & Discussion